Hunting and Spillover in the Amazon region: how economic drivers and extreme climatic events affect wildlife exploitation and emergence of zoonotic diseases

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1 Introduction

Different from plants, which obtain the energy needed for growth by converting light energy into chemical energy through the synthesis of organic molecules in photosynthesis, all animals need to obtain energy for growth and reproduction by eating another living being. Herbivores obtain energy eating plants, and carnivores from other animals, the set of these interactions forms the food web, a complex network involving animals, plants, and microorganisms that channels energy flow and recycles matter, both indispensable for sustaining life. The process through which carnivores obtain food is known as hunting, where they search, kill and eat another animal. Carnivores are know reservoirs of macroparasites (e.g. ticks, fleas, hookworms) and pathogens (e.g. *Lyssavirus* rabies[1], *Toxoplasma gondii*[2], *Brucella* sp.[3]) that can be transmitted to other species by hunting events[4,5].

In a human context, hunting of wild animals to meet food needs was more common prior the development of livestock farming (e.g. cattle, goats, and pigs)[6]. Nowadays, the consumption of wild meat is more restricted to rural populations, indigenous communities, and those worldwide that do not have access to other sources of animal protein[7–10]. The preference for wild meat is often also linked to food security, where communities lacking alternative food sources rely on game animals as substitutes for conventional livestock (i.g., cattle)[11]. However, in urban centers, a paradoxical trend emerges: wild meat consumption increases among wealthy populations, driven by its perceived exotic flavor and status value rather than necessity[12]. Like other carnivores, humans can be infected by several parasites and pathogens during contact with infected animals in hunting events or by consuming wild meat[13–16].

Diseases acquired from animal reservoirs are called zoonoses. This broad term is further divided into three subcategories based on the relationship between the natural reservoir and accidental hosts. Anthropozoonoses describe diseases where animal populations are the original reservoir, but the pathogen can spill over and infect humans upon contact. The opposite scenario defines Zooanthroponoses, where a pathogen that primarily circulates in humans spill over into animal populations. Finally, Amphixenoses are caused by pathogens that move freely and indifferently between human and animal hosts without a strict natural reservoir.[17,18]. For this cross-species jump to succeed, a pathogen must overcome several barriers that act as bottlenecks to transmission (Figure 1). Key among these, are challenges inherent to the pathogen itself, such as its ability to be shed from an infected host, its capacity to survive in the external environment, its capacity to invade and disseminate in hosts of similar or genetic backgrounds, evading their immune systems. Transmission fro one host to the other require efficient transmission modes, from direct to vector-borne or water-borne, to maximize contact with new hosts[19]. Parasites have evolved varied strategies to surmount these obstacles. These include inducing high shedding rates as a stress response, forming protective outer layers—as seen in *Klebsiella pneumoniae*[20]—or demonstrating broad host plasticity, a trait of parasites like *Trypanosoma cruzi*[21]. Furthermore, pathogens can exponentially increase their transmission by incorporating new species into their reproudctive cycle, such as insect vectors (e.g., mosquitoes, sandflies, ticks, flies and fleas)[22–26], or by leveraging the environment itself, as is the case with *Vibrio cholerae*[27] .

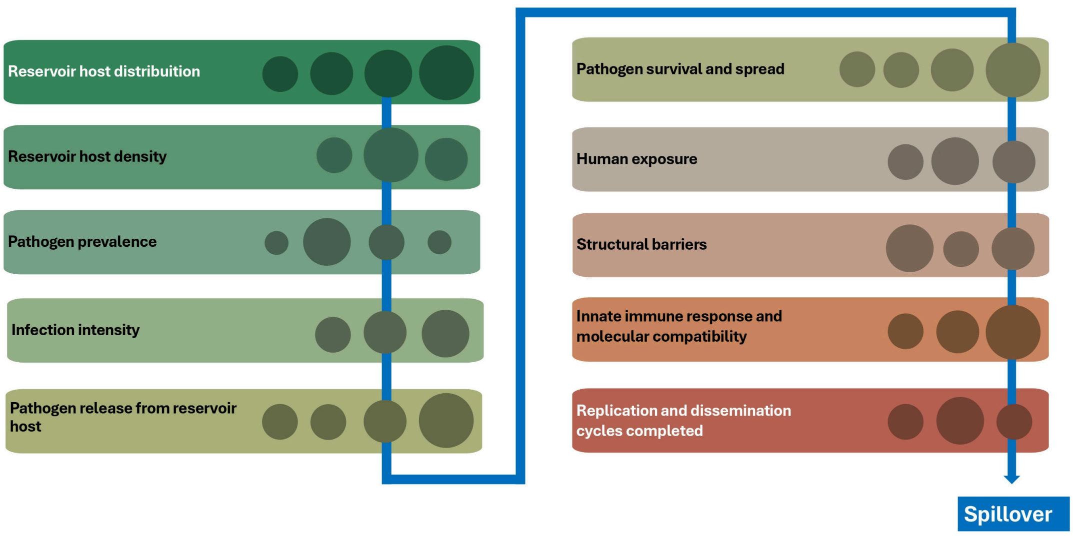

Figure 1: Bottlenecks associated to zoonotic spillover process. Source: Modified from Plowright *et al.*, 2017

Recent threats to global public health such as HIV, Influenza and COVID-19, are Anthropozoonosis. A major historical example is the Human Immunodeficiency Virus (HIV) pandemic; evidence indicates that HIV is a descendant of the Simian Immunodeficiency Virus (SIV), which infects non-human primates[28]. Similarly, recent influenza pandemics have animal origins. For instance, the 2009 H1N1 pandemic virus emerged from a recombination of genetic material from swine, avian, and human influenza viruses[29]. Similarly, COVID-19, which caused a recent global pandemic, is also suspected to have a zoonotic origin. Current evidence suggests it may have emerged from a recombination of bat coronaviruses with another coronavirus from an unknown animal host[30].

Actually, the WHO had a list of pathogens/parasites with pandemic capacity — i.e., COVID-19, Ebola virus, Marburg virus, Lassa Fever, monkeypox, Nipah virus, Henipaviruses, Plague, and Zika virus — indicating a necessity of develop research agenda related to those, identifying transmission pathways, pathogenicity, treatments and control measures[31]. Among these infectious agents, there is evidence that the Nipah virus can spill over to humans during the butchering of bats in Africa[14]. In such scenarios, individuals can accidentally cut themselves, thereby coming into direct contact with contaminated bat blood and tissues, which facilitates transmission.

Rift Valley virus, Nipah virus, and Marburg virus, are anthropozoonoses that can be transmitted during hunting events due to the increased contact between hunters and the hosts[32,33]. On this way, several studies had discussed the risks associated with hunting events itself or meat preparation and consumption of the captured animal, such as bites or scratches during the animal capture or killing, wounds during the butchering, eating uncooked/rare meat, and contact with pathogen inoculum shed by infected animals[10,34–36]. Pathogens are seldom confined to a single transmission route; instead, they often utilize multiple mechanisms, with certain routes dominating their spread. A prime example is *Yersinia pestis*, the causative agent of plague. While its primary mode of transmission is through the bite of an infected flea vector, such as *Xenopsylla cheopis*, it can also spread directly through contact with infected bodily fluids, via airborne respiratory droplets in pneumonic form, or through bites and scratches from an infected animal[37].

2 Hunting in Amazon Biome

Brazilian law prohibits hunting for consumption or and economic purposes, but makes an exception for subsistence hunting in specific contexts involving Indigenous, quilombolas, and some rural communities[38,39]. Permition is granted to such communities that depend primarily on subsistence agriculture, fishing, and hunting for food. However, fishing activities are often seasonally restricted to prevent the overexploitation of fish populations[40,41], a measure that increases their reliance on alternative protein sources, such as wild animals.

Although prohibited, wild meat commercialization occurs in the Amazon region. It is estimated that even in urban regions of Amazon, the annual consumption of wild meat per person can reach 6.5kg[42,43]. This type of estimation are difficult to be made and is probably underestimated, due to its illegal status. Van Vliet made an effort of spent 2 months living in the trifrontier between Brazil, Colombian and Peru region to gain trust by the local people, find the hunters and the people responsible for trade[44].

In the Amazon region, the animals most frequently consumed are *Cuniculus paca*, *Dasyprocta* sp, *Pecari tajacu*, and *Didelphis marsupialis* (Figure 2)[45–49]. Importantly, these species are known reservoirs for zoonotic parasites, including *Echinococcus vogeli*, *Capillaria hepatica*, *Trypanosoma cruzi*, and *Leishmania infantum*[50–54]. These parasites can cause severe and often fatal human diseases. For instance, *T. cruzi* causes Chagas disease, which frequently results in lifelong chronic illness and high mortality rates if left untreated[55]. Other animals widely consumed emcompasses deer, crocodilians, tortoises, primates, and various bird species.

Hunters employ a diverse array of strategies to capture preys. These include hunting during the target animal's active period (e.g., nocturnal foraging), tracking, and the use of tools such as firearms, dogs, traps, bows, and bladed weapons[7,35]. The frequency of hunting expeditions is driven by subsistence needs and can vary greatly, occurring daily, weekly, or monthly. Furthermore, there are reports of extensive hunting trips lasting weeks or even months, after which the captured meat is preserved for later consumption, often by salting[56–58]. In rural communities, not every household has a hunter in the family, but the wild meat are often shared among all members of the community. Meat can also be sold in informal markets[59]

Regarding the determinants of hunting activity in the Amazon region, few studies have focused on the roles of environmental species richness and remoteness from urban centers in driving its occurrence. Based on the existing literature, it is known that hunting pressure is often higher in densely forested areas, which typically support greater species richness and availability of animals (e.g., within protected areas, conservational areas). Conversely, the distance to the nearest large city appears to have a paradoxical effect: while large distances can increase subsistence hunting due to the difficulty of accessing urban markets for alternative protein sources, shorter distances can also elevate hunting rates due to commercial opportunities and access to buyers[8,11,12,46,59,60]

Cuniculus paca	*Dasyprocta punctata*	*Pecari tacaju*	*Didelphis marsupialis*

Figure 2: Main animals hunted or purchased for consumption in the Amazon region

It is important to note that even when alternative protein sources are available, a cultural preference for wild meat often persists, particularly during celebrations and festivals. Beyond nutrition, various animal parts may serve medicinal, ritualistic, or artisanal purposes, as documented in various African communities where animal fats or even feces are utilized in traditional medicinal practices[35,61,62]. Alongside subsistence and cultural drivers, hunting in the Amazon region is also significantly driven by commercial incentives. There are numerous records of various species being hunted for monetary gain, where captured animals are sold in informal markets to supply urban demand for wild meat[42,44,59]. Typically, the meat is sold already butchered and cleaned, though in some areas, such as Iquitos, Peru, live animals are also traded. In Iquitos alone—home to one of the largest wild meat markets in South America —more than 100 tons of wild meat are sold annually. This trade reached a peak of 442 tons in 2018, generating an estimated US$2.6 million in revenue[63]. Despite the prominence of large, accessible markets like that in Iquitos, wild meat is also commonly found in smaller, local fairs throughout the Amazon region. At these venues, wild animals are often sold alongside more common protein sources such as beef, chicken, fish, and vegetables[64]. Recently, during a field expedition, members of our research team were informed of the local sale of *D. marsupialis* at a fair in Cametá (Pará state), a city in the lower Tocantins River basin. In Brazil, wild animals hunting is not concentrated on the Amazon, since Northeast has registered several markets of wild animals for eating, medicinal or even body parts as souvenirs[64,65]

3 Human-mediated Environmental Changes and Potential Impacts on Zoonosis Transmission

3.1 Economic driven environmental changes

The Amazon region stands as one of the most biodiverse areas on the planet, characterized by an enormous abundance of animal and plant species[66]. This exceptional species richness stems from both its vast terrestrial ecosystems and complex aquatic environments[67,68]. This immense natural wealth has supported human societies for millennia, providing essential resources such as food, medicine, and commodities. Subsistence hunting, fishing, and other uses for natural resource and reliance on wildlife by indigenous and local communities are a fundamental aspect of the socio-ecology of the Amazon[47,69]. This intimate association between individuals and the forest ecosystem does, however, create a dynamic interface for the presence and spread of zoonotic diseases. Despite this, outbreaks of zoonotic diseases in the Amazon region have rarely spread outside the region.

However, as such interfaces become more intensified due to larger human interventions and modifications of the environment or urban expansion[70], spillover and large outbreaks become a greater concern.

Large development projects, such as highways, hydropower plants, timber exploitation, and gold mining (specially the illegal one), introduce populations into previously inhabited landscapes[19,70–72], increasing the contact between human and wildlife and raising the potential for the transmission of pathogens and parasites. The construction of highways, while critical for transporting goods and increasing human mobility, also facilitates the spread of parasites and pathogens into regions where they were not previously endemic[73,74]. For example, the Brazil-Peru highway has been associated with an increased burden of dengue fever in proximate urban centers compared to more distant ones[75].

Similarly, the construction of hydroelectric dams raises significant public health and ecological concerns. These projects require extensive landscape alteration, including the creation of large reservoirs that flood vast areas. This transformation severely impacts local ecosystems, disrupting fishing practices, altering animal migration patterns, and changing local flora[76–78]. Crucially, it can also create new dynamics for disease transmission; there are evidence to an increase of malaria incidence following dam construction due to creation of mosquito breeding habitats[79], but such effects appears to have a more complex dynamics associated[80]. Furthermore, the classification of hydroelectric power as an entirely 'clean' or non-polluting energy source is contentious. There is growing evidence that hydroelectric reservoirs, particularly in tropical regions, can be significant sources of greenhouse gas emissions (e.g., methane and carbon dioxide) due to the decomposition of flooded organic matter[81].

The Amazon rubber boom of the 20th century is one of the region's most documented human interventions. It attracted large contingents of laborers to extract latex from *Hevea brasiliensis* for rubber production. This period saw a well-documented increase in hunting activities by the rubber tappers (seringueiros), who relied on wild animals for subsistence while working in remote forest areas[8]. The exploitation of wildlife has also been fueled by subsequent economic cycles. For example, the demand from the global fashion industry in the middle of the 20th century caused a increase in hunting pressure aimed at obtaining the skins and leathers of large felids and crocodilians for the production of luxury items like accessories, handbags, and coats[82,83]. Such interventions are examples of human activities in natural environments that can increase zoonotic (re)emergences[84].

From 1904 to 1969, approximately 23 million animal skins were exported from the Brazilian Amazon, comprising 13.9 million mammal and 7.5 million reptile skins. The jaguar (*Panthera onca*), one of the most valued species, had individual skins sold for US$7.28. Records indicate the commercial slaughter of 804,080 ocelot (*Leopardus pardalis*) and 529,517 margay (*Leopardus wiedii*) skins. This industry operated on such a scale that the city of Manaus alone hosted over 150 major exporters of wild animal skins between the 1920s and 1960s, with commercial empires maintaining fleets to support the trade[85]. As shown in Table 1, municipalities such as Barcelos and Santa Isabel do Rio Negro in the state of Amazonas remain primary hotspots for wildlife traffic.

Table 1. Top 20 municipalities in the Amazon Region associated with traffic of wild animals. Data from Rede Nacional de Combate ao Tráfico de Animais Silvestres report, 2024

Position	Municipality	State	Group of Especies
1º	Barcelos	AM	Ornamental Fish
2º	Santa Isabel Do Rio Negro	AM	Ornamental Fish
3º	Altamira	PA	Ornamental Fish / Birds / Reptiles
4º	Medicilândia	PA	Ornamental Fish / Reptiles
5º	Ilha Do Marajó	PA	Reptiles / Ornamental Fish
6º	Almeirim	PA	Ornamental Fish / Birds / Felines
7º	Serra Do Navio	AP	Amphibians / Birds / Felines / Primates
8º	Manaus	AM	Primates / Felines / Birds
9º	Labrea	AM	Birds / Primates / Reptiles
10º	Itacoatiara	AM	Turtles / Arachnids
11º	Santarém	PA	Ornamental Fish / Arachnids
12º	Gurupi	TO	Birds
13º	Alvorada	TO	Birds / Reptiles
14º	Boa Vista	RR	Birds
15º	Cruzeiro Do Sul	AC	Birds
16º	Caracaraí	RR	Birds / Reptiles
17º	Presidente Figueiredo	AM	Birds / Primates
18º	Porto Velho	RO	Primates / Reptiles
19º	Araguaína	TO	Birds / Reptiles
20º	Colniza	MT	Felines

In 1968, the city of Altamira in the state of Pará produced 36,000 kilograms of wild animal skins for export. The hunt for other species was equally intensive; in 1964, on a single farm in Amapá, over 60,000 ducks (*Amazonetta brasiliensis*) were killed by firegun to supply the fashion industry with feathers[86]. Historical records from the mid-20th century reveal the extensive commercial exploitation of Amazonian manatees (*Trichechus inunguis*), with approximately 140,000 individuals hunted between 1935 and 1954 for their meat and valuable fat[87]. Rather than products made from the animals, the trade now increasingly involves live specimens trafficked for various purposes, as illustrated in Figure 3.

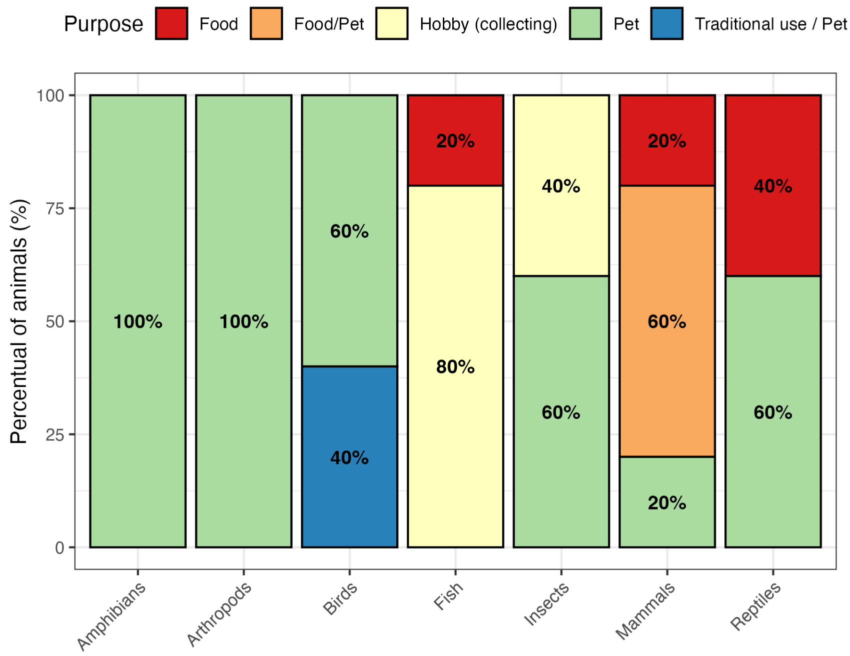

Figure 3: Percentage distribution of the intended use of trafficked species by taxonomic group. Source: Rede Nacional de Combate ao Tráfico de Animais, RENCTAS - Brazil 2024

In addition, land-use modification for agricultural expansion stands as a primary driver of environmental change in the Amazon. The land-use classification for the Brazilian Amazon in 2022, detailed in Figures 4 and 5, reveals a biome increasingly defined by fragmentation. Vast expanses of 'Primary Forest Natural Vegetation' are now interspersed with extensive 'Herbaceous Pasture' and 'Shrub Tree Pasture,' alongside various agricultural crops. Studies indicated that this conversion follows a predictable pattern, with deforestation corridors advancing along transportation networks like roads and rivers. This creates a characteristic pattern of clearing that opens frontier regions to further settlement and resource exploitation[88]. The dominance of pastureland in this converted area is widely understood to be driven by complex economic factors, including commodity cycles and land speculation, establishing a powerful and often self-perpetuating cycle of land conversion.

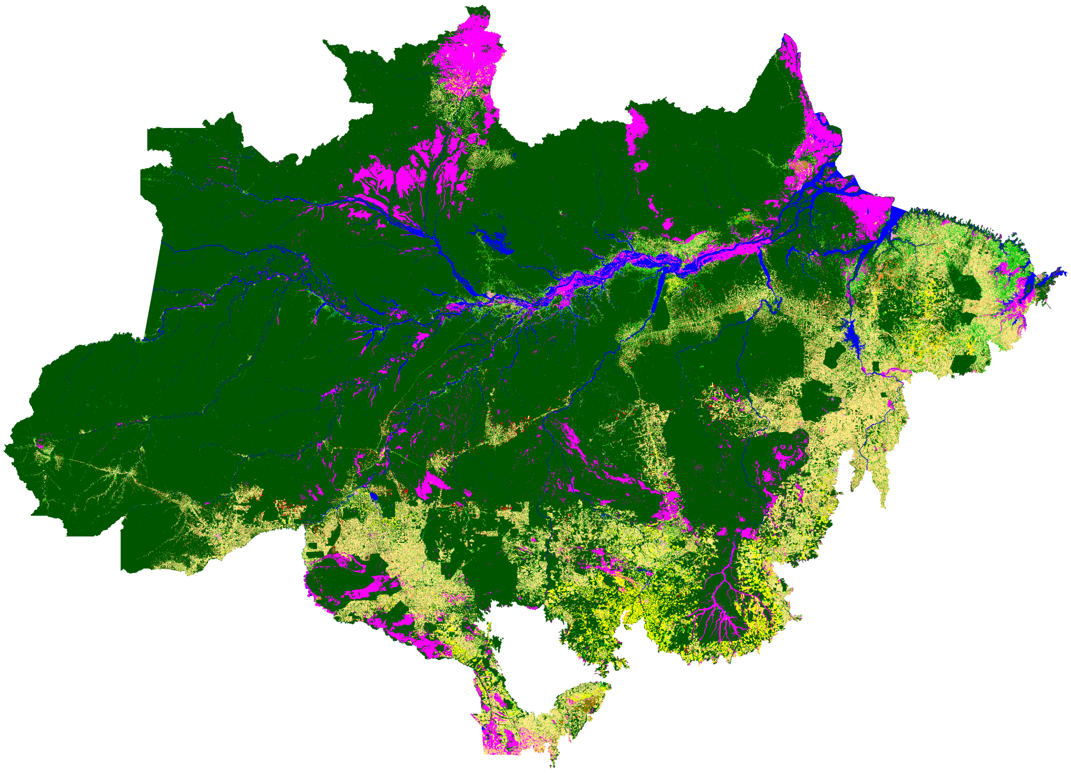

Figure 4: Land-use overview for Brazilian Amazon in 2022. Source: TerraClass Project, Instituto Nacional de Pesquisas Espaciais - Brazil, 2022

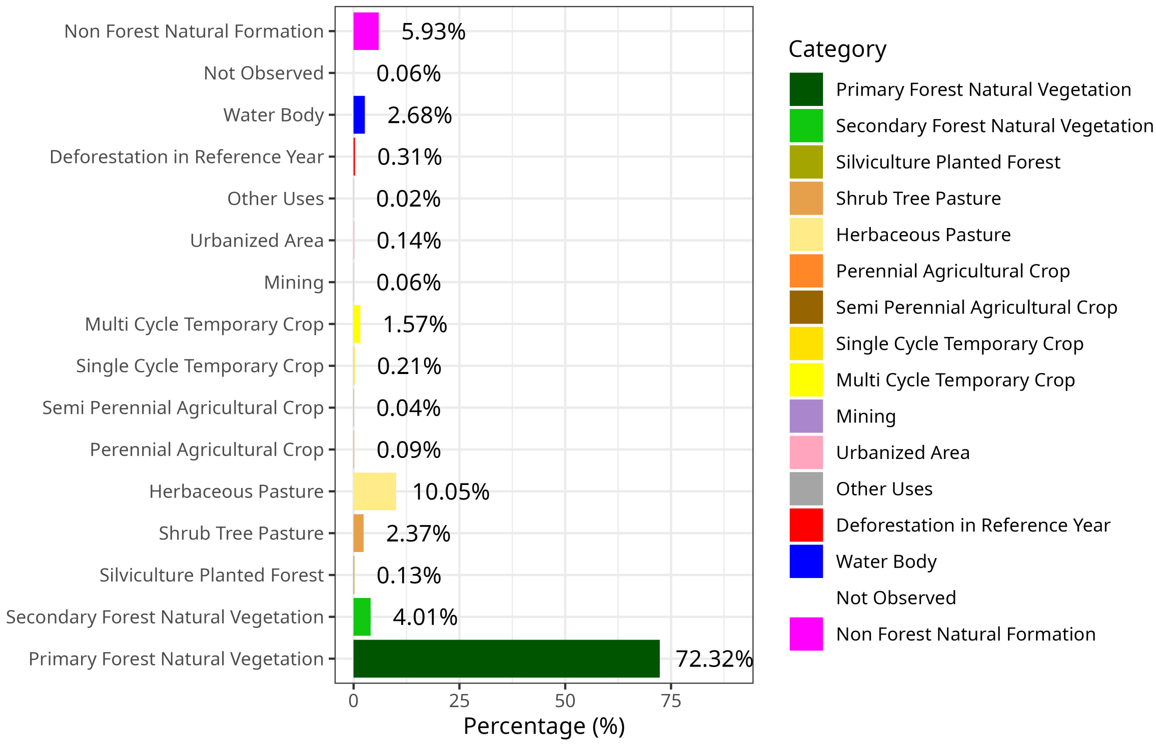

Figure 5: Percentage of each category of Brazilian Amazon Land-Use in 2022. Source: TerraClass Project, Instituto Nacional de Pesquisas Espaciais - Brazil, 2022

The creation of forest edges and the proximity of farmlands to remaining forest patches significantly increase the accessibility of previously remote habitats for hunters and traffickers. This "access effect" has been documented as a key mechanism linking deforestation to wildlife depletion[89]. Moreover, the fragmentation process itself can trigger an "ecosystem decay" of vertebrate communities, reducing population sizes and genetic diversity and thereby rendering species more susceptible to local extinction from overharvesting[90]. Therefore, the same land-use changes that fuel the monetary gains can also establish the precise conditions that can promote transmission of infectious diseases, positioning the degraded agricultural frontier as a potential hotspot for zoonotic events[91].

3.2 Climate change

Human activities are also a dominant force influencing the global climate. The emission of greenhouse gases—largely from fossil fuel combustion, industrial production, agriculture, and large-scale fires—increases the concentration of carbon dioxide and other compounds in the atmosphere[92]. This process heats the planet, contributing to an increased frequency and intensity of extreme climate events such as droughts, floods, and hurricanes. These events have devastating consequences not only for human populations but for the stability of the entire biosphere[93,94].

Consequently, climate change also alters pathogen and parasites transmission dynamics, intensifying or even creating new transmission cycles. A good illustration of these problematic are the transmission of water-borne diseases that followed intense floods, such what happens in regions with low socioeconomic indicators for *Vibrio cholerae*[95,96], and leptospirosis[97,98]. Climate change is also implicated as a key driver in the global spread of vector-borne diseases. Rising temperatures and extreme weather events can expand the geographic range suitable for disease vectors such as mosquitoes and ticks[99,100]. These effects are often non-linear and delayed, as recent research on dengue transmission in Brazil has demonstrated complex relationships following floods and droughts[101].

While the Amazon region has long been defined by a marked climate seasonality—driving annual pulses in river volume, resource availability, and animal behavior[102,103]— a new pattern is emerging. The frequency and intensity of floods and droughts can also lead to changes in human behavior by disrupting fishing activities and impacting riverine communities' mobility, while also promoting high mortality rates for aquatic life[104–106]. This ongoing changes is illustrated by the fact that from 2020 to 2024, the river water level reached historically low drought levels three times in the Amazon region (in 2022, 2023, and 2024) , as seen in Figure 6. This repeated breaking of historical reports has led prominent researchers to posit a permanent shift, or a "new normal," in the region's climate[107,108].

These conditions, coupled with pervasive food insecurity, drive human populations with scarce alternative food resources to intensify their exploitation of wild animals as a primary protein source, increasing hunting activities and the wild meat trade[108,109]. However, animal populations are also severely affected by these events. During droughts, animals are forced to migrate across the landscape to find water and food, often leading them into human settlements in search of resources, and increasing human-animal interactions that can result in a spillover process. During extreme floods, their habitats can be destroyed, resulting in significant animal mortality[110–112]. In both situations, animals are exposed to high levels of stress—from migrating from flooded areas, searching for food, or encountering humans. This stress can induce immunosuppression, increasing the chance of being attacked by an animal, and also their susceptibility to infections[113,114]. Consequently, this leads to a higher prevalence of infected animals and greater spillover risk. The anthropogenic interventions mentioned here and they impacts are summarized in table 2.

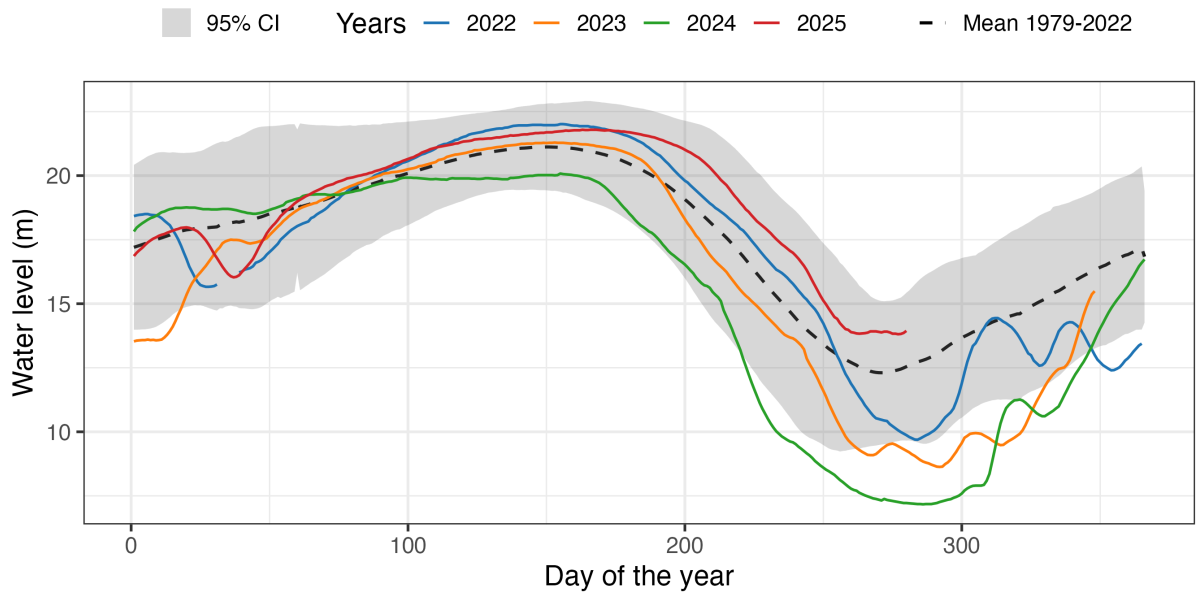

Figure 6: Water level of the Solimões River during dry periods in Fonte Boa, a municipality in the state of Amazonas. Source: Agência Nacional de Águas (ANA) - Brazil, 2025

Table 2. Anthropogenic drivers of disease emergence in the Amazon region.

Intervention Category	Examples	Impacts on Environment and Wildlife	How affect Zoonotic Risk?
Direct Resource Exploitation	• Economic cycles (rubber, skins) • Subsistence hunting • Live animal trafficking	• Drastic depletion of populations • Hunting pressure on target species • Animal stress and immunosuppression	• Increased direct contact between humans and animals • Handling of carcasses and bodily fluids • Exposure during capture and handling
Transport Infrastructure	• Road construction • Highway and river expansion	• Forest fragmentation • Creation of deforestation "frontiers" • Access to remote areas	• Access effect: humans in pristine areas • Pathogen dissemination to new regions

Intervention Category	Examples	Impacts on Environment and Wildlife	How affect Zoonotic Risk?
			• Animals forced into disturbed areas
Energy & Mining Projects	• Hydroelectric dams • Mining (illegal gold mining)	• Radical ecosystem alteration • Displacement of populations • Creation of vector habitats	• Altered transmission dynamics • Creation of mosquito breeding sites • Worker aggregation in forest areas
Land Use Change	• Agricultural expansion • Deforestation and fragmentation	• Forest conversion to pasture • Creation of forest edges • "Ecosystem decay"	• Concentration at agricultural frontier • Biodiversity loss • Hotspot for zoonotic events
Anthropogenic Climate Change	• Greenhouse gas emissions • Large-scale fires and industry	• Extreme droughts and floods • Altered hydrological regime • "New normal" climate	• Behavioral changes in humans and animals • Animal stress and immunosuppression • Geographic expansion of vectors

4 Modelling framework of Hunting-Driven Spillover Risk

To understand the drivers of disease spillover risk mediated by hunting practices, we developed a mathematical framework that represents the interaction between humans, and zoonotic reservoirs through different pathways. The background is a SIS (Susceptible, Infected, and Susceptible) type model that describes the transmission dynamics of a pathogen within a wild animal reservoir and the contact of this reservoir with a human population that leads to spillover events. The human population is divided into two groups: Hunters, H who spend time searching and capturing wild animals, and Consumers, C, who prepare and eat the wild meat (e.g., we consider 3 family members per hunter). The animal population, A, is defined as a mammal species that is hunted and consumed by the human population, and is a reservoir of a particular zoonotic parasite or pathogen (Figure 7). Different transmission pathways, illustrated in a specified section, including injuries during huntings (bites, scratches, cuts), processing transmission (injuries during the meat processing), oral transmission (meat consumption) were considered. We also consider environmental transmission, a more abstract representation of transmission events involving contaminated surfaces with shed parasites (e.g., soil, water), or via vectors, both associated with infected animals. A representation of the model is on the figure below.

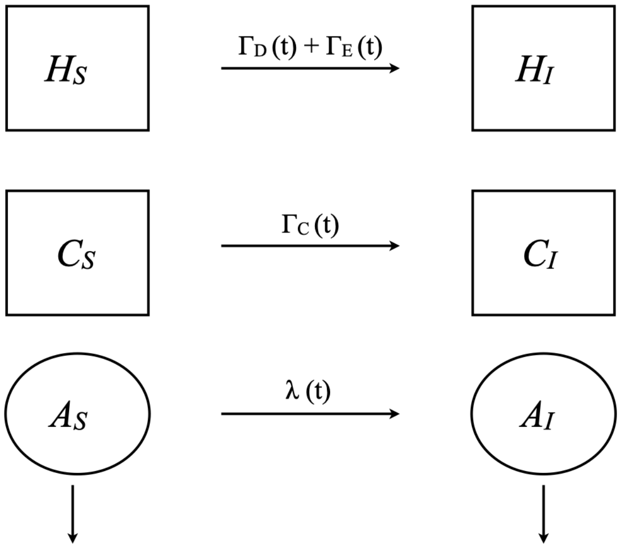

Figure 7: Diagram of compartmental model describing the dynamics of zoonotic spillover from humans

4.1 Population dynamics of the animal population

The population dynamics of an animal species A is governed by a logistic growth curve with an intrisic growth rate r, and an enviromental carrying capacity $K(t)$. Death processes include death from natural causes at a rate μ, and from hunting at a rate $h(t)$ of hunter individuals (H). No additional mortaility by disease is included.

$$\frac{dA}{dt} = rA\left(1 - \frac{A}{K_t}\right) - A(h_t H + \mu) \tag{1}$$

Considering climate seasonality reflected on precipitation as a modulator of humans and animal behavior, represented by precipitation impacting environment, it is assumed to affect the base values of carrying capacity (K_0), intraspecific transmission between animals (λ_0), and hunting efficiency rate (h_0)[112,115–117]. To model these dynamics, we adopted the daily precipitation pattern shown in Figure 8A, which drives the subsequent changes in carrying capacity (Figure 8B), hunting efficiency (Figure 8C), and pathogen transmission rates (Figure 8D).

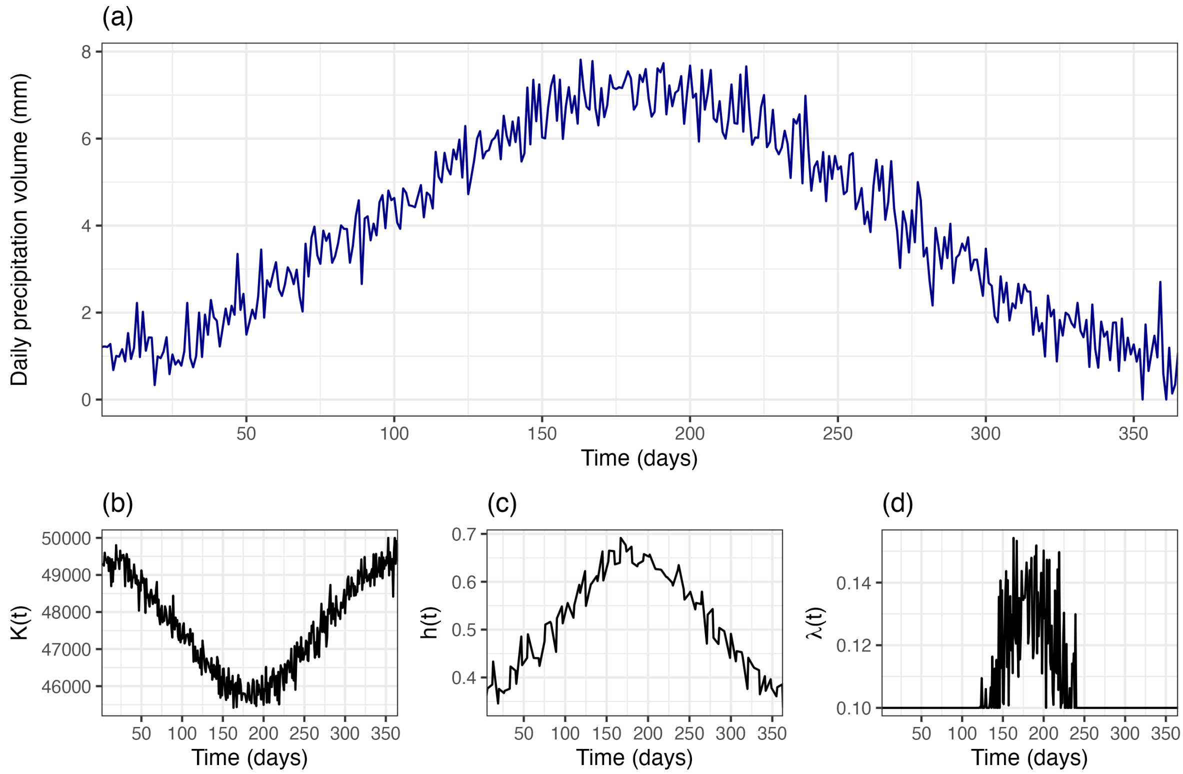

Figure 8: Modeled effects of precipitation-driven climate seasonality. The daily precipitation pattern (A) directly influences three key variables: environmental carrying capacity (B), the hunting efficiency rate (C), and the pathogen transmission rate between animals (D). At the peak of the rainy season, carrying capacity decreases. This forces animals into closer contact, which simultaneously increases pathogen transmission (due to stress and crowding) and hunting efficiency.

4.2 Disease dynamics in the wild population

For simplicity, an SIS model describes the pathogen dynamics within the wild population A, where A_S is the susceptible compartment and A_I, the infected compartment ($A = A_S + A_I$). A_S becomes infected by a given intraespecific transmission rate (λ_t) that vary with seasonality and represents infections during contact between A_S and A_I. Infected ones loses the infection by a recovery rate (η), that represents immunity.

$$\frac{dA_S}{dt} = rA\left(1 - \frac{A}{K_t}\right) - A_S\left(h_t H + \mu + \lambda_t \frac{A_I}{A}\right) + \eta A_I \tag{2}$$

$$\frac{dA_I}{dt} = \lambda_t A_S \frac{A_I}{A} - A_I(h_t H + \mu + \eta) \tag{3}$$

We consider the same daily mortality μ as an indication of an animal that was not clearly sick, which would possibly cause hunters to avoid them. In this model, neither humans or other animal species can transmit pathogen to .

4.3 Probability of spillover to humans

The model considers spillover through different pathways to Hunters (H) and Consumers (C), described in the Figure 8. Hunters can become infected through path$_I$: injuries caused by infected animals during hunting; path$_P$: accidental cuts during meat processing, path$_O$: consumption of raw or undercooked meat, or path$_E$: contacts with shedded parasites or through infected vectors. Consumers, on ther other hand, can only get infected through paths path$_P$ and path$_O$.

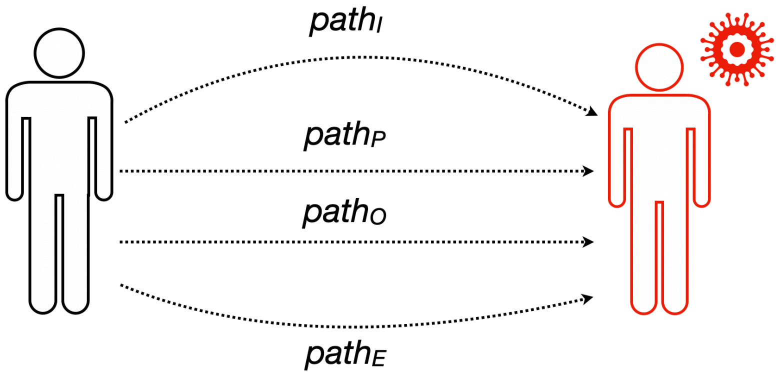

Figure 9: Major transmission pathways of zoonotic spillover

The probability of spillover by each pathway, γ_i, $i = \{I, P, O, E\}$ is modeled as the product of the probability of contact between humans and animals through that pathway l_i and the probability of transmission per contact, p_i. To take in account diferent diseases and contexts, we also included an extra term that weights the relevance of each pathway to the problem at hand, w_i[118].

$$\gamma_i = p_i w_i l_i, \text{where } i \in \{I, P, O, E\} \tag{4}$$

The risk of pathogen spillover to Hunters comprises two categories of transmission: the direct contact risk (Γ_D) and the environmental risk (Γ_E). The first one results from physical interaction with animals and is defined as the sum of the path$_{I-O}$, multiplied by a hunting efficiency rate (h_t). The environmental risk, in contrast, arises from exposure to contaminated habitats, quantified by the path$_E$ parameter multiplied by a general environmental contact rate (ζ). For Consumers (C), the spillover risk is derived exclusively through the direct contact pathway, as their exposure is contingent on the hunters' success. Consequently, the consumer risk is calculated using the path$_P$ and path$_E$ parameters (Γ_C).

$$\Gamma_D(t) = h_t(\gamma_I + \gamma_P + \gamma_O)\frac{A_I}{A} \tag{5}$$

$$\Gamma_E(t) = \zeta\gamma_E\frac{A_I}{A} \tag{6}$$

$$\Gamma_C(t) = h_t(\gamma_P + \gamma_O)\frac{A_I}{A} \tag{7}$$

The equations describing the cummulative incidence of spillover events in the human populations are described below.

$$\frac{dH_S}{dt} = -[\Gamma_D(t) + \Gamma_E(t)]H_S \quad (8)$$

$$\frac{dH_I}{dt} = [\Gamma_D(t) + \Gamma_E(t)]H_S \quad (9)$$

$$\frac{dC_S}{dt} = -\Gamma_C(t)C_S \quad (10)$$

$$\frac{dC_I}{dt} = \Gamma_C(t)C_I \quad (11)$$

4.4 Example of model aplication

Mayaro virus (MAYV) is one of the several arboviruses that threaten human life. Originally described in Trinidad Tobago, where the first cases were identified, this virus was symptoms similar to other arboviruses, such as headache, fever between 39°C and 40.7°C, rash, joint pain, and arthralgia[119,120].

These are related to the difficulty of correct diagnosis of arboviruses in general, which are mostly misdiagnosed as dengue or chikungunya, with recent research estimates that 1% of total dengue notified cases could be in reality caused by MAYV[121]. For these reasons, the pathogen's distribution is under a considerate underestimated. In Brazil, cases are mainly reported in the northern region, especially on Pará, Amazonas, and Rondonia states[122,123].

Cases are related to being accidental spillover, where humans are infected in the sylvatic cycle in rural or forest areas. As other arboviruses, MAYV is a vector-borne disease transmitted between NHP or other mammals, and birds in the sylvatic cycle by *Haemagogus* mosquitoes. However, other species from distinct genera also have vector competence to MAYV (e.g., *Sabethes*, *Coquillettidia*, *Aedes*, etc.)[124]. In addition to that, this virus had a great chance of generating an urban cycle of transmission related to the due the capacity of *Aedes* mosquitoes disseminated it[125].

For our study case, we consider a population similarly to the indigenous community described by Peréz *et al.*, (2019), that has hunting for subsistence as one of the major activities. Transmission would happen during hunting events in forested areas, where hunters have contact with infected *Haemagogus janthinomys* mosquitoes and could be bitten (Figure 10A). In this example, we consider that hunters are searching for one of the MAYV reservoirs, the *Alouatta seniculus* monkey. The community can also be contaminated if mosquitoes with considerable vector competence are present in the village along with infected hunters, but to simplify we only consider the infection of hunters (Figure 10B).

The complete parameters associated to the animal and pathogen, and they references are described in the table 3. Other parameters (Table 4) were arbitrarily defined for carried the simulations, these included the base values of carrying capacity (K_0), and the intraspecific transmission (λ_0)

Table 3. Specific parameters of the animal reservoir and parasite selected for the case study.

Pathogen	Host	r	μ	w	ρ	l	Reference
Mayaro Virus	*Alouatta seniculus*	0.001	0.0001	$w_E = 1$	$\rho_E = 1$	$l_E = 0.49$	126–128

Table 4. Selected model parameters for simulation purpose

Parameter	Description	Unit	Value	Reference
D	Hunting frequency in each week	1	2	Supposed
h_0	Base hunting efficiency rate	1	0.33	7
ζ	Rate of contacts with vector during a hunting event	1	10	Supposed
K_0	Base carrying capacity	Animals	5.000	Supposed
λ_0	Base intraspecific transmission rate	1	0.08	Supposed
η	Animal recovery rate	1	0.1	Supposed

As Mayaro virus (MAYV) is a vector-borne disease, transmitted by infected mosquitoes, and thus considered in the proposed model as an environmental pathway (path_E), the general contact rate (ζ) was parametrized as the average number of *Haemagogus janthinomys* bites received per hunting event of *Alouatta seniculus*. This was made for simplicity, as the model lacks an explicit vector population.

Figure 10: Mayaro virus related to hunting events (A) and to non-hunters (B) considered in the study case

Considering a village with 100 hunters, undertaking two huntings per week, approximately 80% of the hunter population would become infected with MAYV over a ten-year period, even with a relatively low and oscillating animal reservoir prevalence of 4% to 13% (Figure 11). This is a conservative estimate, as it does not account for several key factors in vector-borne disease transmission, such as the extrinsic incubation period, vector feeding habits, or potential protective measures taken by the hunters. Nonetheless, it effectively illustrates the high transmission risk inherent in a single hunting event. In comparison, a seroprevalence study by Pérez *et al.* (2019) reported a population-level seroprevalence of 24% in a sample of 70 individuals. The study further highlighted the elevated risk for households with hunters, which had a seroprevalence probability of 0.68, compared to only 0.21 for households without hunters. While our hypothetical model does not explicitly include transmission to non-hunting consumers, we acknowledge that this occurs in reality, particularly in communities where vectors with substantial vectorial capacity are present (Figure 10B).

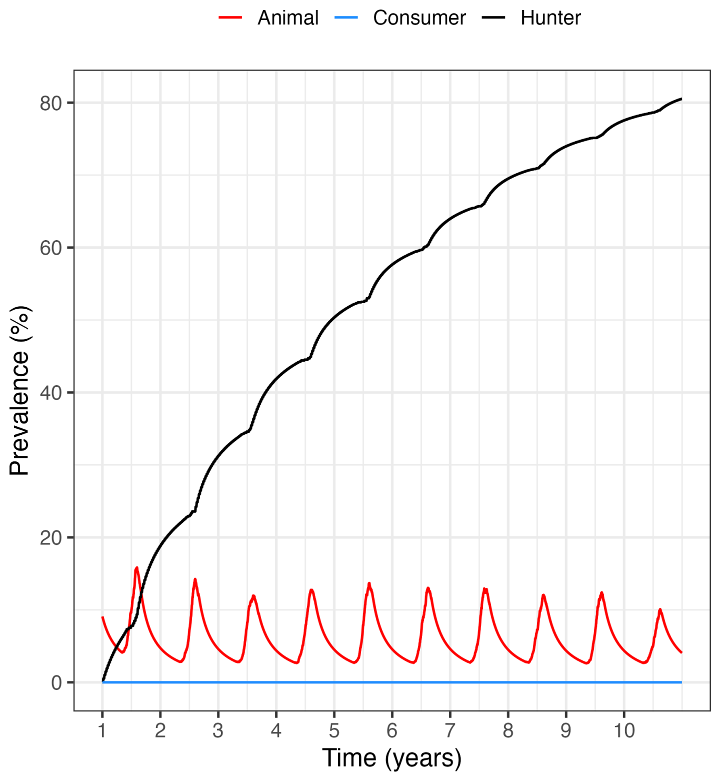

Figure 11: Prevalence of individuals infected with MAYV

We can simulate how transmission would change based on the village's food needs. For instance, we can model scenarios where frequent hunting is necessary—more than twice a week—and compare them to scenarios where hunting is less critical, occurring only once a week or every fortnight (Figure 12A). The model shows a direct correlation between hunting frequency and MAYV prevalence. In a scenario of low hunting intensity, only approximately 30% of hunters become infected. However, as the frequency of hunting events increases, infection rates rise sharply, with simulations showing nearly the entire hunter population becoming infected under conditions of high intensity.

A further investigation was conducted by varying the total number of hunters in conjunction with the frequency of hunting events. This combined analysis revealed a critical, counterintuitive interaction: when the hunter population is increased threefold, the model predicts a decrease in disease prevalence, even when hunting frequency remains high (Figure 12B). This phenomenon occurs because the larger number of hunters leads to a rapid depletion of the animal reservoir. The intensified hunting pressure decimates the host population, which in turn reduces the overall force of infection and limits successful transmission to hunters, despite their increased exposure in the forest.

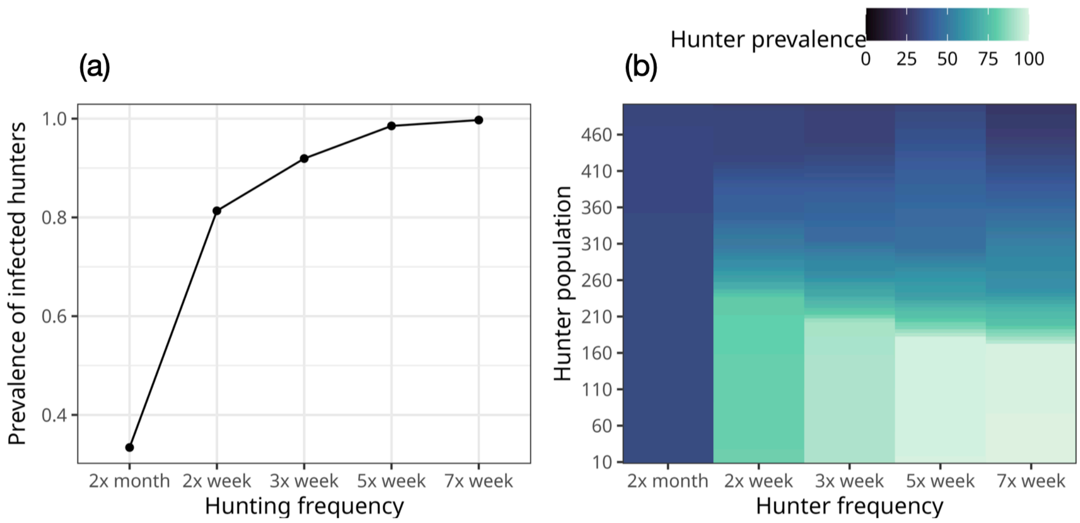

Figure 12: Investigating the effect of hunting events frequency isolated (A) and also varying the total number of hunters (B)

We similarly investigated the impact of other critical parameters, namely the hunting efficiency rate (h_0) and the base intraspecific transmission rate (λ_0). The latter proved to be highly sensitive, as even small increases in its value resulted in a high prevalence of infected hunters (Figure 13A). In contrast, increasing the hunting efficiency rate (h_0) produced an effect analogous to increasing the total number of hunters: it led to a decrease in the number of infected hunters (Figure 13B). This is because higher efficiency accelerates the depletion of the animal reservoir (even with low number of hunters), thereby diminishing the impact of a high intraspecific transmission rate.

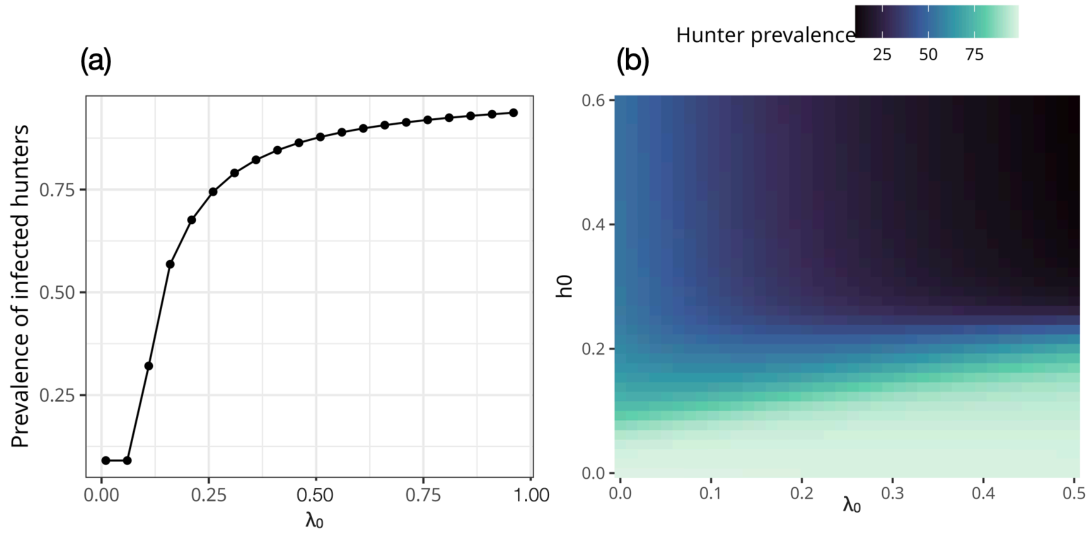

Figure 13: Investigating the effect of intraspecific transmission (λ_0) isolated (A) and also varying the hunting efficiency rate (h_0)

These modeled scenarios can manifest in reality due to various environmental pressures. Extreme climate events, such as severe droughts or floods, alongside anthropogenic habitat destruction, can force animal populations into confined areas. This crowding effect significantly promotes the intraspecific transmission of MAYV. Furthermore, these same events can devastate the subsistence agriculture of local communities, creating food insecurity that compels increased hunting frequency and efficiency. Thus, environmental disruptions create a dual-pressure system: they intensify the transmission of the virus within the reservoir population while simultaneously driving greater human exposure to it, significantly elevating the risk of spillover.

5 Conclusion

Our economic activities in the Amazon such as infrastructure projects, resource extraction, agricultural expansion, historical trade in wildlife, and human-induced climate change have profoundly altered the region's ecology These actions disrupts ecosystems, reduce the variety of life, and push people and wildlife closer together. This increased contact raises the risk of diseases transmission between animals and humans. Together, these anthropogenic interventions converge to create ideal conditions for the spread of pathogens from animals to humans.

Despite its limitations, the model presented here theoretically demonstrates, in quantitative terms, how hunting activities can facilitate the zoonotic spillover. Although spillover events may be rare at the individual level, the risk for people who regularly interacts or consume wild meat is not zero. Furthermore, recent epidemics and pandemics underscore the profound global risk posed by the emergence of a novel pathogen, highlighting that the consequences of such spillover events can be catastrophic.

The model proposed here have several limitations as taking in account the hunting of a unique species at time, instead of the convenience-hunting in which hunters capture animals that have the opportunity. This aimed-hunting in a long-term leads to a complete exploitation of the animal populations, which is a very unrealistic situation by two main reasons: 1. rural and indigenous populations not feed only on wild animals, and 2. specially with a small population of hunters used here. The assumption that hunting occurs only two days per week is a conservative estimate. In economic-driven hunting situations the frequency of huntings can be greater. In this context, several considerations are relevant. An increased volume of hunting for profit would not only raise the transmission risk for the hunters themselves but also for the entire commercial trade chain. Critically, this amplified activity could enable pathogens to transcend geographic barriers, potentially introducing them into previously unaffected urban centers.

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